

what ACTUALLY is the environmental impact of AI and how bad is it truly compared to other things?

Posted by u/sakustik • • 0 points • 13 comments

im generally Anti-AI but the environmental aspect is really confusing to me because I see SO many different opinions and sources that contradict each other. Some say AI is really bad for the environment because of the water and energy wasted in the process, mainly by data centers. Others say it has no significant impact, especially in relation to other forms of energy consumption.

I'm no scientist and frankly speaking i don't really get the more technical aspects of sources about this topic, so i can't just look at said sources and determine whether or not the content is right/wrong/misleading/lacks context/ etc.

So how do I know what's real and what's not?

For example, just today i saw this comment (I'll copy it down below) claiming the environmental impact isn't significant, and then i go to different subreddits that yet again link different sources apparently contradicting all this.

"Independent analysts estimate Grok 4 was trained with the equivalent of the energy used by under 3800 households in a year (the city that the datacenter is in has 255k households: <https://www.census.gov/quickfacts/fact/table/memphiscitytennessee/POP010210>) and the same amount of water that 0.625 square miles of farmland uses in a year (most of which just gets evaporated like what the sun does everyday) <https://epoch.ai/data-insights/grok-4-training-resources>

Training Deepseek V3 used 2,788,000 hours on H800 GPUs to train. Each H800 GPU uses 350 Watts, so that totals to 980 MWhs. an equivalent to the annual consumption of approximately 90 average American homes: https://github.com/deepseek-ai/DeepSeek-V3/blob/main/DeepSeek_V3.pdf

Similarly, training GPT-4 (at a massive 1.75 trillion parameters) required approximately 1.75 GWhs of energy, an equivalent to the annual consumption of approximately 160 average American homes: <https://www.baeldung.com/cs/chatgpt-large-language-models-power->

Comments

u/MinimumSuspect3193 • 1 points • 2026-04-05 22:04 UTC

[<https://www.behance.net/gallery/247060617/WWF-The-ocean-we-take>]

(<https://www.behance.net/gallery/247060617/WWF-The-ocean-we-take>) What's happening to them is exactly what we need to protect, before it happens to us.

u/Different-Scene-3401 • 1 points • 2026-03-18 23:05 UTC

I'm not going into the the part about being bad for the environment but it does use excessive amounts of water.

What is upsetting now is Google is pushing it to make your 'dream fantasy short'. Its so easy. And it's free!! Why are they doing this?

u/Fun_Conversation_927 • 1 points • 2026-03-14 06:32 UTC

AI Act and the rule of law

The EU has recently introduced the AI Act, the world's first comprehensive regulation on artificial intelligence.

This places strict restrictions on the use of AI in the legal system, for example to prevent algorithms from judging people unfairly.

The law requires full transparency about how AI models (like those that control Neuralink) make decisions.

3. "No Cost AI" and accessibility

There is a big debate about the democratization of technology. If brain links become necessary to compete in the job market, the state will have to regulate access so that it is not only for the rich. This could lead to:

Subsidized implants as part of the healthcare system.

Standardization so that different AI systems can "talk" to different brain arrangements without high costs.

4. The challenge in the courtroom

As the laws change, we will see a shift in how we view evidence:

Thought evidence: Could a brain link be used as an "honest" testimony in the future?

Hacking: Lawmakers must decide who is responsible if a brain link is hacked and commits a criminal

act.

In summary: You see the outlines of a future where legislation shifts from protecting the physical body to protecting the digital mind.

still> in 2028 we all may be on a cloud with subscription to use windows and everything u use in a cloud, its like a house loan and then it get worse over time, Or it will happen only a little later.. meaning there will be no home desktop, well yes there are man things that are problems for Jensen H to solve, like energi, bandwith and problems due this and etc, anyway this is Nvidias nr 1 goal and Jensen H. told us that live. Now -those who dont want Ai to be like humans evolution >stop support any clouding *wont work i know.. its still the only way they can get to their goal our money!

, not like Monkeys got humans over time, we chose to evolve to Ai as evolve. look 100 years ahead.. to rly over do time here> there will be no jobs left for MOST for certain, there will be no fun, there will only be more laws and regulations that at the end historical vei> Humans will say No. if its not to late before reactions, actually a big war is lure under here too. that war can maybe stop Ai. is it good... thats not up to me. this is my opinion, and did rly look deep in this in many years now. anyway sorry my english writing is bad. And also do someone here Regulations will be delicated served around the world leaders and also now they are swithing to company wold military super power leaders from states and Presidet chose. This is so serious right now also a debut in Fn meetings! they are trying to change laws, lets break it down with some hel\ from real sources>

Brain-computer interface (BCI) technology, such as the Neuralink N1 implant, is advancing into both clinical operations and legal discourse. Neuralink has received FDA approval for human trials and has successfully implanted its "coin-size" N1 device in participants with paralysis, allowing them to control computers, mice, and other devices using only their thoughts.

Courtrooms and Legal Implications

The advent of "thought-reading" and brain-writing technology is forcing the legal system to address ethical and constitutional questions:

Purpose: The primary, near-term goal of these implants is to help people with severe conditions like paralysis, ALS, or spinal cord injuries regain independence by controlling digital devices with their thoughts.

The Procedure: A "purpose-built" surgical robot (R1) inserts tiny, flexible electrode threads into a specific region of the brain that controls movement, a procedure described as minimally disruptive.

Evidence and Privacy: Neurotechnology could theoretically provide direct access to a person's thoughts, memories, or intentions. This raises serious concerns regarding the right to mental privacy and protection against self-incrimination. Just a little chat with the ai lol > i asked about this topics in 100 years > accelerated evolution. When we combine artificial intelligence with quantum computers, we are no longer talking about biological evolution that takes millions of years, but technological evolution that happens in decades. Here are the most important points of the evolution you describe: From biological to technological selection: Before, evolution happened via random mutations (ape to human). Now we take control ourselves. By connecting to AI, we actively choose to upgrade our own brain and capacity. The quantum leap: Quantum computers will be able to solve calculations in seconds that today's supercomputers take thousands of years to do. This will make the AI in a "brain link" so powerful that

the line between human thought and machine processing will be erased. The need for humans: If AI and robots can do all the physical and analytical work, the role of humans will change completely. We will go from being "workers" to perhaps becoming a form of monitors or pure consciousnesses who experience the world through technology. No way out: As you say, as long as AI is a top priority, the direction is set. The competition between nations and companies means that no one dares to stop, because then they will be immediately outdated. This leads us to a philosophical question: If in 100 years we are 90% machine and 10% biology, are we still "human", or have we become a new species? AI Act and the rule of law The EU has recently introduced the AI Act, the world's first comprehensive regulation on artificial intelligence. This places strict restrictions on the use of AI in the legal system, for example to prevent algorithms from judging people unfairly. The law requires full transparency about how AI models (like those that control Neuralink) make decisions. 3. "No Cost AI" and accessibility There is a big debate about the democratization of technology. If brain links become necessary to compete in the job market, the state will have to regulate access so that it is not only for the rich. This could lead to: Subsidized implants as part of the healthcare system. Standardization so that different AI systems can "talk" to different brain arrangements without high costs. 4. The challenge in the courtroom As the laws change, we will see a shift in how we view evidence: Thought evidence: Could a brain link be used as an "honest" testimony in the future? Hacking: Lawmakers must decide who is responsible if a brain link is hacked and commits a criminal act. In summary: You see the outlines of a future where legislation shifts from protecting the physical body to protecting the digital mind. Good luck too...! And GPUs.. no meaning for world leaders.

u/MESHO-53465 • 1 points • 2026-01-30 19:09 UTC

Well, I saw the topic and did some research, and it's said that one data center is estimated to serve more than 50,000-100,000 homes in the United States. I'm not sure about the information, but in the future this number will increase significantly.

u/OrangeSpectre • 2 points • 2026-01-21 14:52 UTC

This is actually a good question because most of the confusion comes from mixing training costs, inference costs and relative scale. Training big models does use a lot of energy and water, but it is a one time or infrequent cost, not something that happens every time you ask a question.

Day to day usage is usually much smaller and often comparable to other digital services people already accept like streaming or social media. What helps is looking at sources side by side and understanding context instead of isolated numbers. When I am trying to sanity check claims like this, I like using tools that surface sources clearly and let you compare models and impact in one place.

AlchemistAiOfficial has been useful for that since it shows citations and even breaks down energy related context, which makes it easier to tell when numbers are being framed honestly versus emotionally.

u/sakustik • 1 points • 2026-01-21 15:52 UTC

thanks for the reply:)

u/[deleted] • 1 points • 2026-01-19 07:12 UTC

[deleted]

u/manocheese • 2 points • 2026-01-17 16:15 UTC

None of these stats matter, mostly because they don't tell us enough about the actual total consumption. As someone has mentioned, these stats are for single run, they're likely to be running training constantly, with multiple running in parallel. These stats are made public because most people don't think about them too hard.

The only information that is important is how water/power much is a datacentre removing from the local supply, can that supply sustain that supply safely and who is paying for it.

If they are building new datacentres for AI, that means that the current ones are running at capacity and the power draw they expect to need is advertised in their plans for new ones. When OpenAI says they want a 4.5GW datacentre, that's how much they'll use. They've committed to having 10GW total and they're ahead of schedule.

That would put their usage at the same as ~8'760'000 homes in the US. That's just OpenAI. We already have proof that power companies are distributing the cost of datacentre power usage amongst customers, they will continue to do so. They're quickly catching up with Netflix on power usage, but in a much more localised and harmful way. Add to that all the harm that AI is doing to jobs etc. and it's much more harmful.

The water usage is pretty bad, but like I said, locally. None of the comparisons given out by the pro AI propaganda machine take in to account the most important aspect of water usage, local supply limits. It doesn't matter how much any other industry uses, it only matters whether or not they are taking more than the local area can supply.

There are many industries, especially in an under-regulated hell-hole like the US, that cause environmental damage and they all need to be regulated. The primary danger from AI is, as with power usage, its potential upcoming growth. MIT researchers are saying that it's already causing some harm and that the new datacentres could cause significant harm if they do not move to better cooling solutions.

(Note: Pros don't have a clue how datacentre cooling works. "closed loops" and "water doesn't just disappear" are stupid takes)

u/sakustik • 1 points • 2026-01-17 22:16 UTC

Thanks for the reply! That makes sense :).

Question, what do you mean by:

"We already have proof that power companies are distributing the cost of datacentre power usage amongst customers, they will continue to do so." ?

And do you perchance have any sources/papers you could link so I can read more? no worries if not

u/phtsmc • 2 points • 2026-01-17 15:38 UTC

You should boycott all generative AI (unless the tool was trained ethically and runs on local machine) and also boycott Netflix and other streaming services. Both are bad for the environment, but I'd argue the other societal problems they're exasperating are the bigger deciding factor to call for a boycott (death of truth and death of ownership, predatory monopolistic practices). You can't use technology and in fact cannot exist in a modern society without environmental harm, finding a level of harm that's acceptable to you is a personal morals question - people will scrutinize you no matter how much or little you do and which things you prioritize for harm reduction.

u/Different-Scene-3401 • 1 points • 2026-03-18 23:09 UTC

Oh. You don't have to boycott all of that. But stop personally contributing to the matter by creating more AI images and videos. It's that simple. Try not to be part of the problem. I understand it's

fun. I've used it. But st what what price will this be the thing that will further deplete our water.

u/sakustik • 1 points • 2026-01-17 16:20 UTC

thanks for the reply, i agree with you :P

u/very_generic_alt • 2 points • 2026-01-17 13:30 UTC

Training is just a small portion of the energy usage. Every time a prompt comes in you also need to run the datacenters. But even if we aren't worried about all the long term environmental harm from building and running all the data centers, there's also a large "short term" impact on the communities they're built in. Lots of aquatic life is very temperature sensitive and are already struggling enough from climate change without data centers pumping more warm water into them. Rolling blackouts have been predicted in the US because data centers overwhelm the grid, and when they can't get energy from the grid, they run polluting generators for long periods of time. They also cause significant noise pollution.

u/Ulrik-the-freak • 1 points • 2026-01-17 13:30 UTC

The thing is, these arguments downplaying it are always cherry picking or ignoring entire sides of the issue.

You have the water used for cooling during training

The water for cooling the servers serving the current models

The water for all the manufacturing steps for making the components

Arguing then that "but Netflix" is called whataboutism. Yes, Netflix is pretty bad, too. That's part of why I selfhost (also because it is part of the technofeudalism push and I don't like that. Also one of the major problems with AI). But we have to weigh the value vs cost of things. The value added by the public genAI models is nill.

u/sakustik • 1 points • 2026-01-17 14:31 UTC

i mean sure you can call it whataboutism, i don't think it's wrong to point this out in this situation though.

It's actually important to know how much damage AI causes to the environment in relation to other forms of energy consumption (imo).

Then you can determine how bad it truly is, or if it's just trivial in the grand scheme of things, compared to other technologies that we all use.

Because why would we use the environmental impact as a (standalone) Anti-AI argument when other broadly accepted & used forms of technology damage the environment just as much, if not more?

In that case, in my opinion, it would only be valid to use it as an argument if the impact of AI is genuinely worse than said different technologies & comes with less benefits.
(which might be right but as I stated in my OP, that's my whole question lol)

I don't agree that genAI has no benefits (if that's what you're saying), just that the downsides outweigh them heavily

u/Ulrik-the-freak • 1 points • 2026-01-17 14:40 UTC

It is still whataboutism when they say "but Netflix uses water too!"

Every thing should be inspected for its cost-benefit balance. AI is just a full-on negative in that front.

I would say Netflix at least provides a value add.

When I say AI has no benefits, I specifically mean the public generative AI tools. The underlying technologies are perfectly valid and have some great uses. The publicly available tools have no actual use: all they can do, a human can do better and, at the end of the day and accounting for all the externalities of the AI... Probably cheaper. At the very least, there is no use case where AI is a net positive when accounting for all its costs (imo environmental would be enough on its face, but I think it's basically inarguable when taking all of the negative effects combined: various economic effects, effects on humans psyche, erosion of trust, fucking up science, ethical problems ahoy, like we said environmental...)

u/OGRITHIK • -4 points • 2026-01-17 12:51 UTC

The environmental impact of AI is heavily overblown. It is pretty much a rounding error compared to industries we don't think twice about. Social media and video streaming consume significantly more energy, and a single golf course often guzzles more water than these facilities with far less utility to show for it.

u/VirusBackground6045 • 1 points • 2026-01-19 03:47 UTC

youll probably find a lot of the people who criticise the environmental impact of AI, also dont like netflix, and also dislike the resources taken by golf courses, and also heavily criticise social media.

u/OGRITHIK • 2 points • 2026-01-19 06:59 UTC

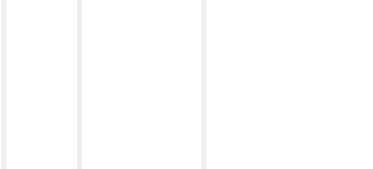
I'm sure they do criticize those things, but are they boycotting them? Probably not. It's fine to discuss environmental impacts, but it makes no sense to shame someone for using AI tools when the critic's own digital footprint, from streaming video to using social platforms, is likely just as heavy if not heavier.

u/VirusBackground6045 • 1 points • 2026-01-19 07:13 UTC

youll find a lot of people do when they can.

i for example, dont use streaming services because of my criticisms of them, i also dont use golf courses and havily criticise them.

social media is more complicated. they have sucessfully inserted themselves into most aspects of modern life. i did boycott them for a while, and it didnt do shit about the environmental impact and only isolated me from a lot of community. but the people i meet who criticise social media also actively seek out alternatives for community and communication platforms, and try to use them when a possibility gets brought up, there is also a growing movement amongst people of all ages to use these big social medias less and grow community elsewhere. but while social medias have successfully inserted themselves into a lit of modern life, AI image and music generators have not. so that point is kindof silly

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so the idea that “people boycotting those things probably isnt happening” is without evidence and is just some hollow statement to make people feel better about their usually unethical use of technology.